

Adroit Technologies

Protocol Driver

Name	Beckhoff TwinCAT ADS-DLL API
DLL	BECKHOFF



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1. Overview

For this driver to work correctly the machine must have a running version of the Beckhoff's TwinCAT Server. The Driver Supports TwinCAT2 and TwinCAT3. Please ensure it is installed.

See [Appendix A](#) for TwinCAT2 Installation and [Appendix B](#) for TwinCAT3 installation.

1.1. Required Files

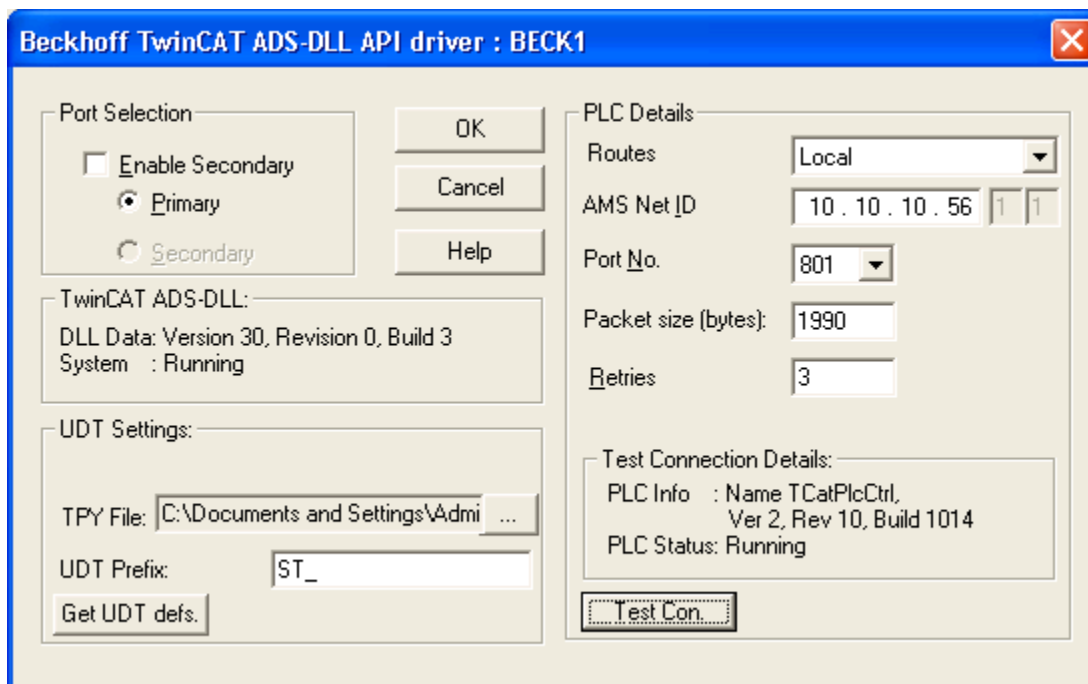
This driver was developed using AdsDll.dll version 28, revision 0, build 14. This file or newer must be present in the Windows path for the driver to be configured or run.

2. Wiring

Not applicable.

3. Device Configuration

3.1. Device Configuration



3.1.1. Port Selection

Enable Secondary – Enables secondary configuration for dual redundant PLCs via the **primary** and **secondary** radio buttons. Do not enable if a secondary channel is not configured correctly.

3.1.2. UDT Settings

TPY File. – The TPY File is a file from the Beckhoff System that contains all the UDT Data. The Driver will import this file and automatically create all the UDT's as specified in this TPY file. Click on the browse button to select a file to open, and then press the "Get UDT Defs." Button to create all the UDT's. A window will open after the creation process that will show all the UDT's created and allow for manual intervention, if required.

UDT Prefix – Prefix for UDT's to search for in the TPY File.

Get UDT defs. – This button will be enabled as soon as the "Test Con." Action has succeeded. This will open the TPY File specified and automatically create all UDT's that can be found in the file with the specified Prefix.



PLEASE NOTE

From Adroit 7.0.10.0 onwards, you will have a Browse functionality on the scan dialog that will read all the symbols from the TPY file into the browser and allow you to select the symbol. This will then construct the scan address accordingly.

3.1.3. PLC Details

Routes – The Routes allow selection of Remote Beckhoff Units or just the remote. These routes will be read from the registry where Beckhoff saves its configured routes. If communication is to be on the local machine, just select "Local". Also, "Custom" is available whereby the user can then manually specify the AMS Net ID in case of the driver failing to fetch the routes from the registry as in the case of TwinCAT3.

AMS Net ID – The IP Address of the device you wish to communicate to. The default IP Address will be the hosting PC's IP Address. The last two digits are fixed.

Port No. – A drop down list of available supported ports.

Packet size (bytes) – The maximum number of bytes to read in one job. The maximum available is 1990 bytes and a minimum of 80 bytes (default string length).

Retries – The number of times that Adroit will try to gain communication with a PLC before failing a transaction.

The driver uses a fixed timeout of 5000 milliseconds on transactions.

3.1.4. Test Connection Details

Test Con. – A 'test connection' button is provided and will report the details of the type of PLC and its status.

4. Supported Addresses

Device	Min Range	Max Range	PLC Data Type	Boolean	Integer	Real	String
Inputs I (read only)	0	65535.8	BOOL	IX:0.0			
		65534	WORD		IW:0	IW:0	
Outputs Q	0	65535.8	BOOL	QX:0.0			
		65534	WORD		QW:0	QW:0	
Internal M	0	65535.8	BOOL	MX:0.0			
		65535	Byte		MB:0		
		65534	WORD		MW:0	MW:0	
			Int		MW:0 S	MW:0 S	
		65532	DWORD		MD:0	MD:0	
			Dint		MD:0 S	MD:0 S	
			Float (Real)			MD:0 F	
		65534	String				MW:0.80 C
UDT's	See Below						

5. Protocol Definition and Considerations

5.1. Scanning UDT's

To Scan any UDT, just follow the following structure:

UDT:<ProgramName>.<MyUDT> <StructureName>

Where:

<ProgramName> = If the UDT Instance is in the Global Variable Scope of the Beckhoff Software, then this stays empty, but if it is any other place than Global Variables, then the scope must be specified, like MAIN as example.

<MyUDT> = The Structure Instance as created in Beckhoff software.

<StructureName> = UDT(Structure) Name as created in the Beckhoff software.

Example(MAIN Program Scope): UDT:MAIN.myTest1 ST_DEMO

Example(Global Variable Scope): UDT:.myTest1 ST_DEMO

UDT's can be setup for callbacks instead of polling. To do this, just change the scan rate to 0 when scanning. This is for the purpose of timestamping. To enable timestamping for a UDT, you must create 2 DWORD slots called "**FILETIME_HIGH_DWORD**" and "**FILETIME_LOW_DWORD**", which must be packed with the FILETIME's high and low 32-bit words.

The driver also support ENUMS and UDT's in UDT's.



PLEASE NOTE

This driver does not support Arrays in UDT's.

5.2. Scanning VARIABLES by Name

To Scan any Variable by its name, just follow the following structure:

VAR: <ProgramName> .<VarName> <VarType>

Where:

<ProgramName> = If the Variable Instance is in the Global Variable Scope of the Beckhoff Software, then this stays empty, but if it is any other place than Global Variables, then the scope must be specified, like MAIN as example!

<VarName> = The variable name as specified in Beckhoff. Software

<VarType> = Variable Type, such as BOOL,BYTE,INT,DWORD etc..

Example(MAIN Program Scope): VAR:MAIN.myVar1 DWORD

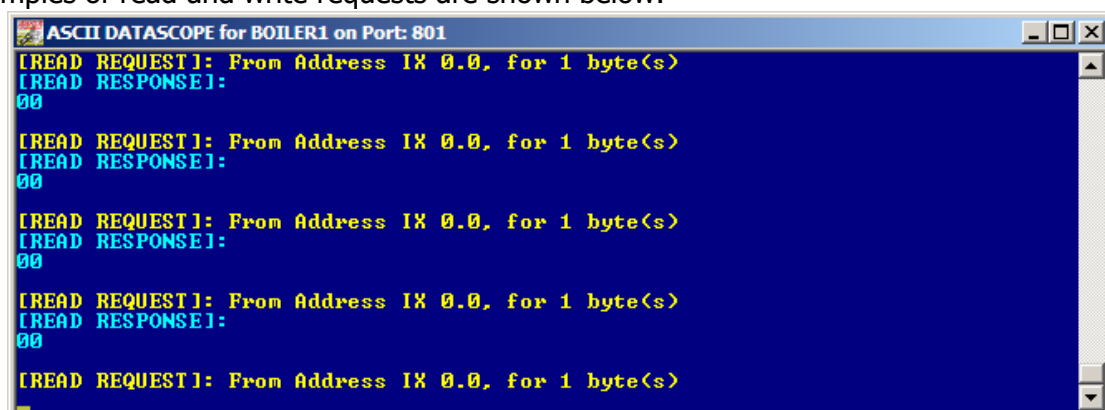
Example(Global Variable Scope): VAR:.myVar1 DWORD

Adroit Variable Types MAPPED to Beckhoff Variable Types

Adroit Variables	Beckhoff Variables
BOOL	BOOL
BYTE	BYTE USINT SINT
WORD	WORD UINT
DFLOAT	REAL
Not Supported	TIME
Not Supported	LREAL
DINT	DINT LONG
DWORD	UDINT DWORD
INT	INT SHORT

5.3. Message Protocol

Examples of read and write requests are shown below:



```

ASCII DATASCOPE for BOILER1 on Port: 801
[READ REQUEST]: From Address IX 0.0, for 1 byte(s)
[READ RESPONSE]:
00

[READ REQUEST]: From Address IX 0.0, for 1 byte(s)
[READ RESPONSE]:
00

[READ REQUEST]: From Address IX 0.0, for 1 byte(s)
[READ RESPONSE]:
00

[READ REQUEST]: From Address IX 0.0, for 1 byte(s)
[READ RESPONSE]:
00

[READ REQUEST]: From Address IX 0.0, for 1 byte(s)
[READ RESPONSE]:
00

```

```

ASCII DATASCOPE for BOILER1 on Port: 800
[READ REQUEST]: From Address QX 2.0, for 1 byte(s)
[READ RESPONSE]:
00

[WRITE]: To QX 0.2, Value: 1
[WRITE SUCCESSFUL...]

```

Example of writing to and reading from a string:

```

ASCII DATASCOPE for BOILER1 on Port: 800

[WRITE]: To MW 10, Value: Hello World
[WRITE SUCCESSFUL...]

[READ REQUEST]: From Address MW 10, for 80 byte(s)
[READ RESPONSE]:
48 65 6C 6C 6F 20 57 6F 72 6C 64 00 CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD
CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD
D CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD CD

```

Errors will be displayed as follows on the error log screen; errors are also event-logged:

```

ERROR LOG for BOILER1 on Port: 300
[READ ERROR]: ...Timeout elapsed - Driver fail on read.
[READ ERROR]: ...Device is not in a ready state - Driver fail on read.
[READ ERROR]: ...Device is not in a ready state - Driver fail on read.
[READ ERROR]: ...Device is not in a ready state - Driver fail on read.
[READ ERROR]: ...Device is not in a ready state - Driver fail on read.
Comms error, cannot read from device - Driver failed.
[READ ERROR]: ...Device is not in a ready state - Driver fail on read.
[READ ERROR]: ...Device is not in a ready state - Driver fail on read.
[READ ERROR]: ...Device is not in a ready state - Driver fail on read.
[READ ERROR]: ...Device is not in a ready state - Driver fail on read.
[READ ERROR]: ...Timeout elapsed - Driver fail on read.
[READ ERROR]: ...Timeout elapsed - Driver fail on read.
[READ ERROR]: ...Timeout elapsed - Driver fail on read.
Successful read from device - Driver healthy.
Primary port Initialized, Secondary port Initialized - Device healthy
[READ ERROR]: ...Timeout elapsed - Driver fail on read.
[READ ERROR]: ...Timeout elapsed - Driver fail on read.
[READ ERROR]: ...Timeout elapsed - Driver fail on read.
[READ ERROR]: ...Timeout elapsed - Driver fail on read.
Successful read from device - Driver healthy.

```

When writing to and reading from PLC Strings:

Keep in mind, that when writing to and reading from a String in the PLC, a maximum of 80 characters is allowed (Adroit's maximum string length), this includes a terminating NULL value at the end of the string, so in essence only 79 of the 80 characters are read/writable.

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

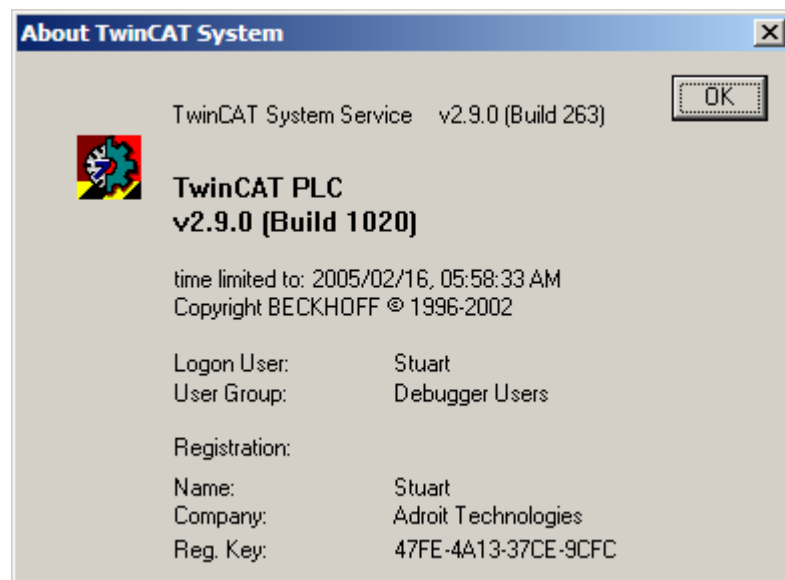
Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

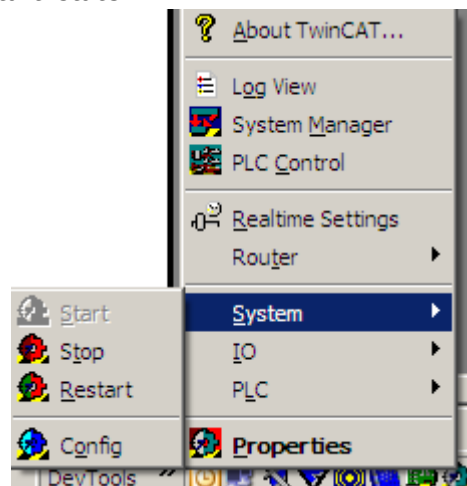
For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7. Appendix A – TwinCAT2 Installation



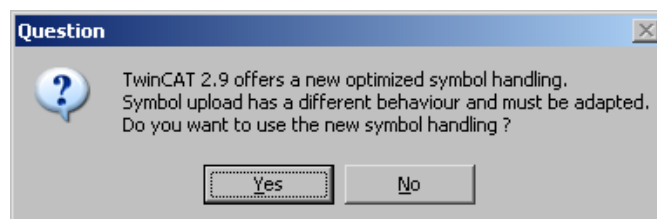
Ensure that the system is in a 'start' state



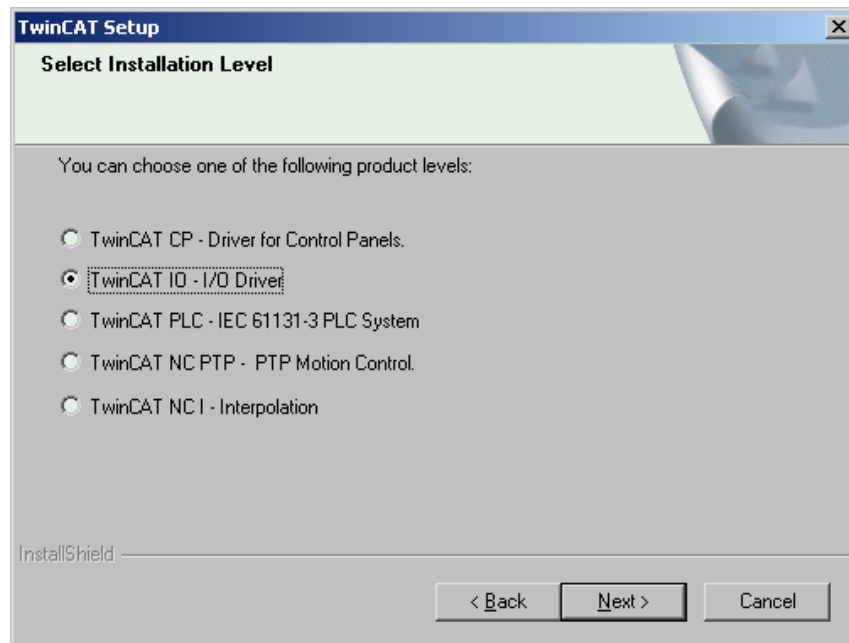
Alternatively, you can install CP driver. This is obtainable from your Beckhoff distributors.

7.1. TwinCAT Server Installation

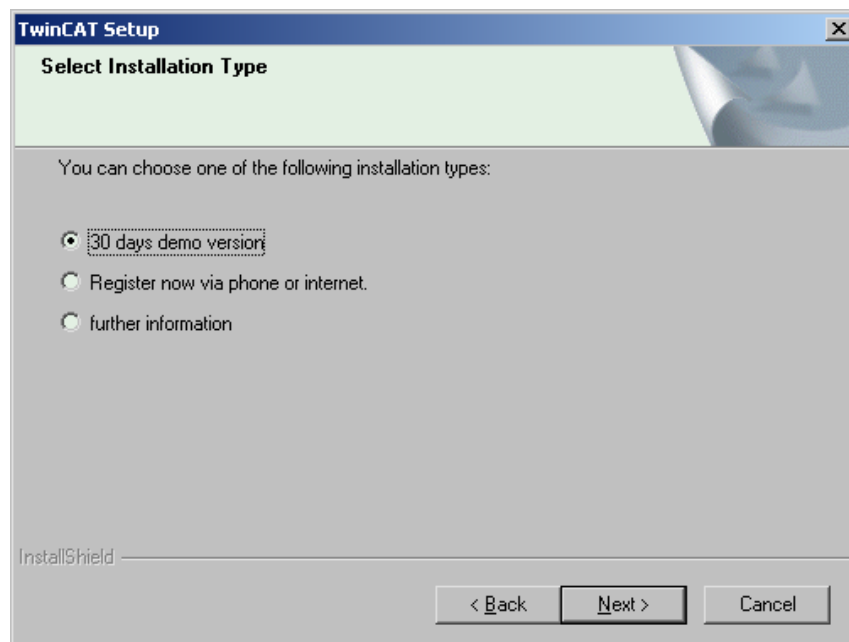
During the installation of the TwinCAT server, the install program may present some install options as below.



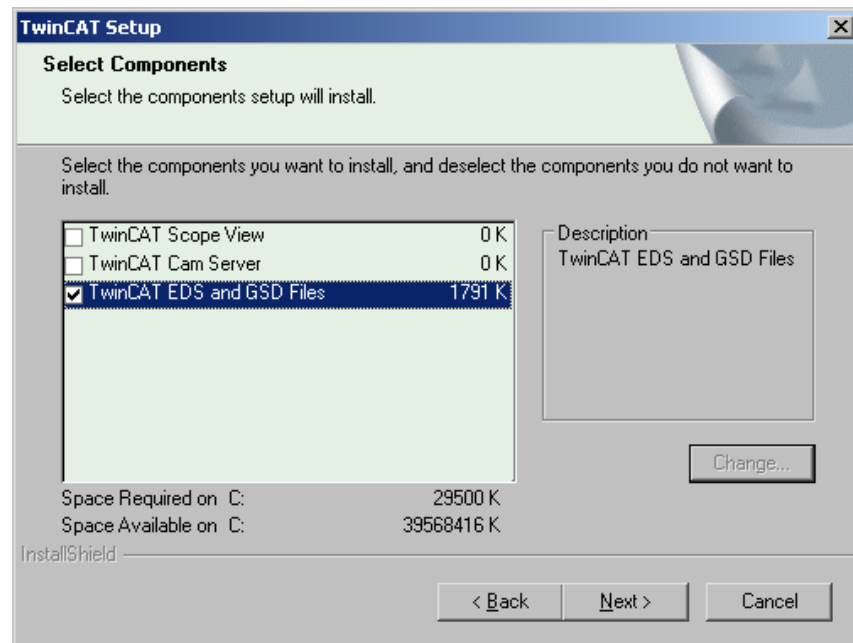
This setting for symbol handling will not affect the Adroit driver.



Select the “TwinCAT IO I/O driver” at this point as shown.



You may select the 30 day demo but will have to register the product latter; please obtain assistance from your Beckhoff distributor to register the product.

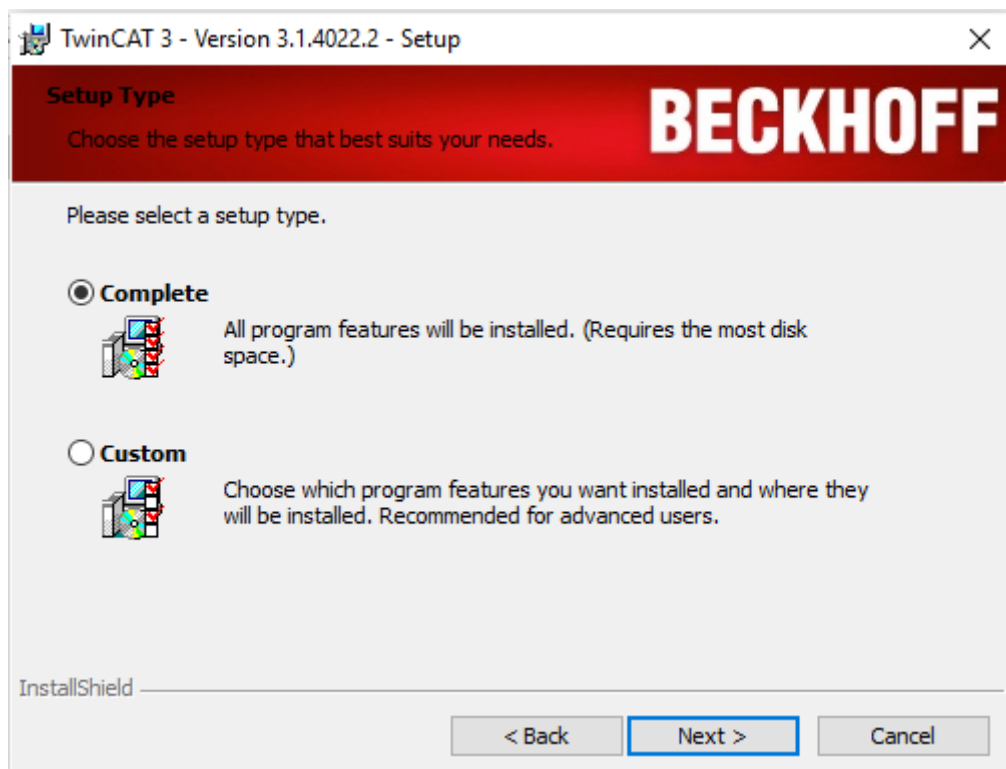


We selected the EDS files for installation simply to ensure that all module catalogues would be up to date.

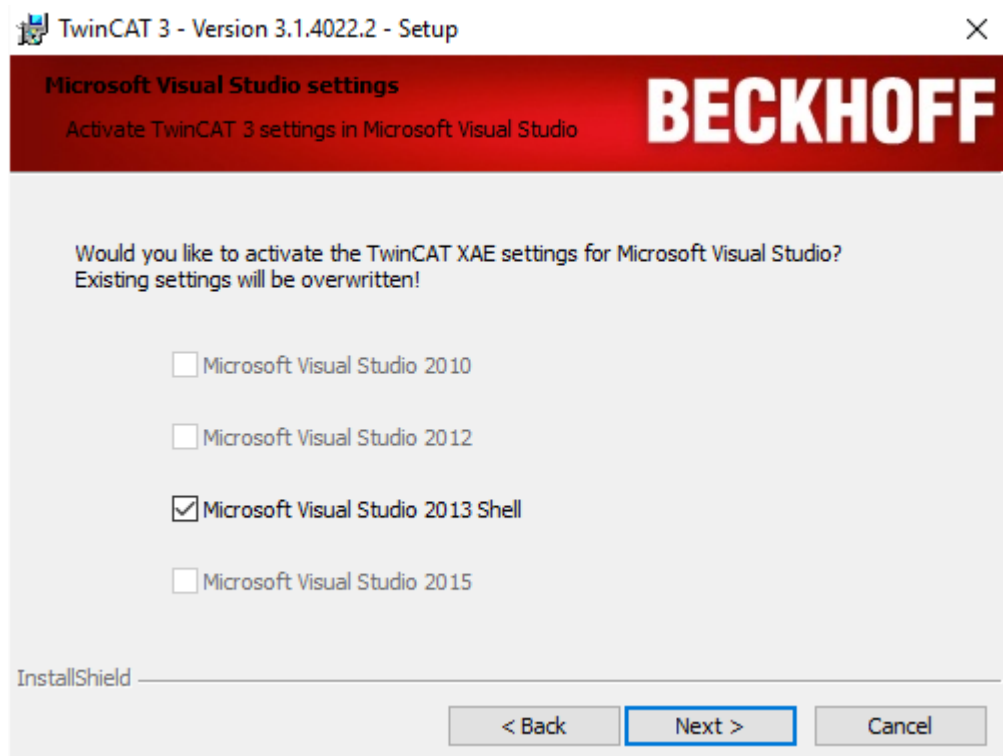
8. Appendix B – TwinCAT3 Installation



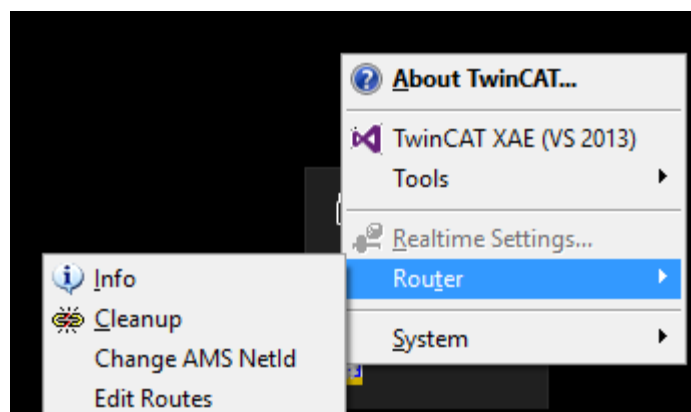
Click Next. Accept the agreement and enter user details.



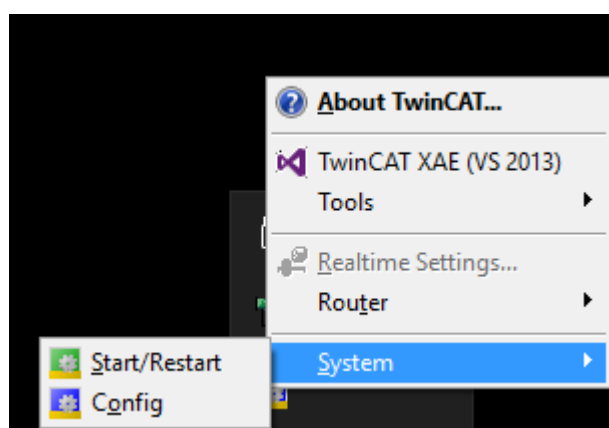
Select Complete and press Next.



Select the Visual Studio 2013 Shell if no other Visual Studio is installed. Press Next and Then Install.



After Install ensure to configure Remote Routes if required by selecting "Edit Routes".



Ensure the system is started.

9. Revisions

9.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.2	19-Jan-2022	K. Robinson	Added links to appendix, changed image alignments.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

9.2. DLL Revision

Rev.	Date	Changes
	31/01/2005	First Release of Driver.
	09/03/2005	Added 'Real' PLC data type conversion for DWORDS
	12/08/2005	TwinCAT server installation covered.
10	30/10/2009	New Additions to driver
11	12/02/2010	Added UDT inside UDT Support
	15/05/2012	Added Variable Type Mapping and new Browser functionality
	18/09/2017	Added TwinCAT3 Functionality.
27	12/04/2022	Fixed AdsCallbackHandler
28	18/04/2023	Fixed agent server load crash if scanning jobs with gaps in indexes i.e. scan QW:X and QW:X+34, the job could except if protected memory was encountered in memory between QW:X+1 -> QW:X+33 as the previous buffer for these would not yet exist. Job sizing was incorrect if scanning negatively extended jobs could cause a heap corruption fixed.

